

Aman Preet Singh Gulati

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Experience

Software Development Engineer II

Noida, UP

Hekaos

Oct 2024–Present

- Designed and developed **scalable distributed systems** with **agentic workflows** automating business processes, reducing manual intervention by 70%.
- Built production-grade **RAG (Retrieval-Augmented Generation) pipeline** using **LangChain** and **OpenAI GPT-4** for enterprise chatbot, enabling intelligent query resolution across 10,000+ HR documents and company policies with 95% accuracy.
- Implemented **vector database** integration with semantic search capabilities, reducing information retrieval time from 5 minutes to 10 seconds.
- Developed AI-powered automation workflows integrating **LLM APIs** to optimize system performance and reduce response latency by 60%.

System Engineer

Noida, UP

Tata Consultancy Services Ltd.

April 2023–Oct 2024

- Led backend development for **cloud migration** of enterprise legacy reporting system serving Fortune 500 pharmaceutical client with 9,000+ retail locations, processing 500,000+ daily healthcare transactions.
- Architected **microservices-based reporting engine** using **Spring Boot**, achieving 40% performance improvement through optimized database queries and caching strategies.
- Designed **RESTful APIs** with **Spring Data JPA** enabling real-time data synchronization across distributed systems with 99.5% reliability.
- Implemented **OAuth 2.0** and **JWT authentication** using **Spring Security**, ensuring HIPAA compliance for protected health information (PHI).
- Integrated **SonarQube** achieving 90%+ code coverage and reducing critical vulnerabilities by 85%, managing multi-environment deployments using **Spring Profiles**.

Associate System Engineer

Noida, UP

Tata Consultancy Services Ltd.

Jul 2021–March 2023

- Developed **Spring Boot microservices** leveraging **Java 8+ features** (Streams, Lambda, Optional) enhancing code performance and maintainability by 30%.
- Engineered complex database operations using **Spring Data JPA** and **Hibernate**, implementing custom queries and indexing strategies to optimize query performance.
- Delivered 20+ feature enhancements and API endpoints in bi-weekly sprints, maintaining 98% on-time delivery rate.

Projects

Enterprise AI Chatbot with RAG Pipeline

- Built **conversational AI system** using **RAG architecture** providing instant access to organizational knowledge including HR policies, employee handbooks, and compliance guidelines.
- Implemented **document processing pipeline** using **LangChain** and **OpenAI embeddings** to vectorize and index 10,000+ documents into **Pinecone vector database**.
- Achieved 95% accuracy in answering employee queries with **semantic search engine**, reducing HR support tickets by 45%.
- Tech Stack:** Python, LangChain, OpenAI GPT-4, Pinecone, FastAPI, PostgreSQL, Redis.

Healthcare Data Analytics and Reporting Platform

- Architected **microservices-based reporting platform** processing 500,000+ daily transactions across 9,000+ retail locations with real-time analytics.

- Developed **RESTful APIs** for data aggregation and visualization, enabling business intelligence dashboards for tracking KPIs and operational metrics.
- Implemented distributed caching and database optimization reducing report generation time from 45 seconds to 8 seconds (82% improvement).
- Built automated data validation pipeline ensuring HIPAA compliance, processing 2M+ records monthly.
- **Tech Stack:** Java 11, Spring Boot, Spring Data JPA, PostgreSQL, Redis, Apache Kafka, Maven.

Project Portfolio Management System

- Engineered **enterprise project tracking platform** enabling delivery managers to monitor 100+ concurrent projects and optimize resource allocation.
- Developed **RESTful API suite** with 30+ endpoints for project lifecycle management, financial tracking, and automated reporting with role-based access control.
- Implemented real-time dashboard APIs providing instant visibility into project health metrics, improving decision-making speed by 50%.
- **Tech Stack:** Java, Spring Boot, Spring Data JPA, Spring Security, MySQL, Hibernate, Maven, Swagger.

Education

Sastra University, Chennai <i>Master of Computer Application</i>	<i>Aug 2022 – July 2024</i>
Dr. Virendra Swarup Institute of Computer Studies (VSICS) <i>Bachelor of Computer Applications</i>	<i>Aug 2018 – May 2021</i>

Skills and Technologies

Programming Languages: Java, Python, SQL, JavaScript
Backend Frameworks: Spring Boot, Spring Security, Spring Data JPA, Hibernate, FastAPI, LangChain
AI/ML Technologies: LangChain, OpenAI GPT-4, RAG Architecture, Vector Databases, Pinecone, LLM Integration
Databases: PostgreSQL, MySQL, MongoDB, Redis, Pinecone
Tools and IDE: IntelliJ IDEA, Git, GitHub, SonarQube, Postman, Swagger, Maven, JIRA
APIs and Integration: REST APIs, OAuth 2.0, JWT, Apache Kafka

Certifications

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- Microsoft Azure Fundamentals (**AZ-900, AI-900, DP-900**)
 - Microsoft Azure Developer Associate (**AZ-204**)